```
[pgslave]
host2 ansible_host=192.168.122.113 ansible_connection=ssh
ansible_user=centos ansible_password=centos123

[all:children]
pgmaster
pgslave
```

The *host1* and *host2* entries are added in the */etc/hosts* file on the host system as shown below:

```
192.168.122.174 host1
192.168.122.113 host2
```

You can test connectivity from Ansible to the CentOS guest VMs using the following Ansible commands:

```
$ ansible -i inventory/kvm/inventory pgmaster -m ping
host1 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}

$ ansible -i inventory/kvm/inventory pgslave -m ping
host2 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
```

The Common setup

The PostgreSQL server needs to be installed on both the master and slave instances, and the database needs to be initialised on both. The Postgres user password is changed. Although the password is listed as a variable in the playbook, it can be encrypted and stored using Ansible Vault when used in production. The *firewalld* daemon is started and port 5432 for the database is allowed through the firewall. The playbook to install and configure the base PostgreSQL server is given below:

```
---
- name: Common pre-requisites
  hosts: all
  become: yes
  become_method: sudo
  gather_facts: yes
  tags: [common]

  vars:
    db_password: "postgres123"

  tasks:
    - name: Install pgdg-centos96 RPM
      package:
        name: https://yum.postgresql.org/9.6/redhat/rhel-
7-x86_64/pgdg-centos96-9.6-3.noarch.rpm
        state: present

    - name: Install PostgreSQL RPM
      package:
        name: "{{ item }}"
        state: latest
      with_items:
        - "postgresql96-server"
        - "postgresql96-contrib"
        - "nano"

    - name: Initialize database
      shell: sudo ./postgresql96-setup initdb
      args:
        chdir: /usr/pgsql-9.6/bin/

    - name: Start PostgreSQL server
      systemd:
        name: postgresql-9.6
        enabled: yes
        state: started

    - name: Wait for server to start
      wait_for:
        port: 5432

    - name: Change postgres password
      shell: sudo -u postgres psql -c "ALTER USER postgres
WITH password '{{ db_password }}'"

    - name: Start and enable firewalld
      systemd:
        name: firewalld
        enabled: yes
        state: started

    - name: Allow postgresql through firewall
      firewalld:
        service: postgresql
        permanent: yes
        state: enabled

    - name: Reload firewalld
      shell: firewall-cmd --reload

    - name: List reloaded firewall
      shell: firewall-cmd --list-all
```

The above playbook can be invoked using the following command:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/postgresql.yml --tags common -vv --K
```